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・ 对称ランダムウォーク

def : 对称ランダムウォーク

$$X_j = \begin{cases} 1 & (w_j = H) \\ -1 & (w_j = T) \end{cases}$$

$$M_0 = 0 \quad \text{or} \quad \text{or}$$

$$M_k = \sum_{j=1}^k X_j \quad , \quad k = 1, 2, \dots$$

○ 期待値, 分散

$$E[X_j] = \frac{1}{2} (1) + \frac{1}{2} (-1) = 0$$

$$\text{Var}[X_j] = \frac{1}{2} (1)^2 + \frac{1}{2} (-1)^2 = 1$$

For

$$E[M_k] = 0$$

$$\text{Var}[M_k] = k.$$

◦ 独立增量

$$M_{k_{i+1}} - M_{k_i} = \sum_{j=k_i+1}^{k_{i+1}} X_j$$

$$\text{Var}[M_{k_{i+1}} - M_{k_i}] = k_{i+1} - k_i$$

◦ 随机游动

$$E[M_L | \mathcal{F}_k]$$

$$= E[M_L - M_k | \mathcal{F}_k] + M_k$$

$$= 0 + M_k = M_k.$$

◦ 2: R 変分.

$$\begin{aligned} \Sigma[M, M]_k &\equiv \sum_{j=1}^k (M_j - M_{j-1})^2 \\ &= \sum_{j=1}^k \left\{ \sum_{l=1}^j X_l - \sum_{m=1}^{j-1} X_m \right\}^2 \\ &= \sum_{j=1}^k \left\{ \cancel{X_1} + \cancel{X_2} + \dots + \cancel{X_{j-1}} + X_j \right. \\ &\quad \left. - (\cancel{X_1} + \cancel{X_2} + \dots + \cancel{X_{j-1}}) \right\}^2 \\ &= \sum_{j=1}^k X_j^2 = k \left(\neq \text{Var}[\Sigma[M]_k] \right) \end{aligned}$$

↑
実際に計算してる？

・ 大数の法則と中心極限定理

$$\overline{W}_t^{(n)} \equiv \frac{1}{\sqrt{n}} M_{nt}$$

○ 期待値 分散

$$E[\overline{W}_t^{(n)}] = 0$$

$$\text{Var}[\overline{W}_t^{(n)}] = \frac{1}{n} (nt) = t$$

○ マルコフ性

$$E[\overline{W}_t^{(n)} | \mathcal{F}_s] = \overline{W}_s^{(n)}$$

o 2nd part Remark $W_t^{(n)} = \frac{1}{\sqrt{n}} M_{nt}$

$$[W^{(n)}, W^{(n)}](t)$$

$$= \sum_{j=1}^{nt} \left(\frac{W_j^{(n)}}{n} - \frac{W_{j-1}^{(n)}}{n} \right)^2$$

$$= \sum_{j=1}^{nt} \left(\frac{1}{\sqrt{n}} M_j - \frac{1}{\sqrt{n}} M_{j-1} \right)^2$$

$$= \frac{1}{n} \sum_{j=1}^{nt} \left(\cancel{M_1} + \cancel{M_2} + \dots + \cancel{M_{nt-1}} + M_{nt} - (M_0 + \cancel{M_1} + \dots + \cancel{M_{nt-1}}) \right)^2$$

$$= \frac{1}{n} \sum_{j=1}^{nt} (X_{nt})^2$$

$$= t$$

Th 3.2.1 中心極限定理)

$$\overline{W}_t^{(n)} \xrightarrow[n \rightarrow \infty]{d} N(0, t)$$

$$\frac{1}{\sqrt{t}} \overline{W}_t^{(n)} \xrightarrow[n \rightarrow \infty]{d} N(0, 1)$$

$$\therefore) E[e^{\theta \overline{W}_t^{(n)}}]$$

$$= E\left[\exp\left\{\theta\left(\frac{1}{\sqrt{n}} \sum_{j=1}^{n+1} X_j\right)\right\}\right]$$

$$= E\left[\frac{n+1}{1!} \exp\left(\frac{\theta}{\sqrt{n}} X_j\right)\right]$$

$$\Leftarrow \exp\left(\frac{\alpha}{n} X_1 + \frac{\alpha}{n} X_2\right)$$

$$= \exp\left(\frac{\alpha}{n} X_1\right) \exp\left(\frac{\alpha}{n} X_2\right)$$

$$= E \left[\exp\left(\frac{\theta}{\sqrt{n}} X_1\right) \exp\left(\frac{\theta}{\sqrt{n}} X_2\right) \dots \right]$$

$$= \prod_{j=1}^{n+1} E \left[\exp\left(\frac{\theta}{\sqrt{n}} X_j\right) \right]$$

∴ 1.

$$\left(E \left[e^{\theta X_i} \right] = \frac{1}{2} e^{\theta} + \frac{1}{2} e^{-\theta} \right)$$

$$= \prod_{j=1}^{n+1} \frac{1}{2} \left(e^{\frac{\theta}{\sqrt{n}}} + e^{-\frac{\theta}{\sqrt{n}}} \right)$$

$$= \left\{ \frac{1}{2} \left(e^{\frac{\theta}{\sqrt{n}}} + e^{-\frac{\theta}{\sqrt{n}}} \right) \right\}^{n+1} = \varphi_n(\theta)$$

$$\lim_{n \rightarrow \infty} \varphi_n(\theta) = \exp\left(\frac{\theta^2}{2}\right)$$

∴ 4. の 証明 終 了 . に つ づ け

$(x \sim \frac{1}{\sqrt{n}})$

$$= \frac{1}{2^2} \log \left(\frac{e^{2\theta}}{2} + \frac{e^{-2\theta}}{2} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \varphi_n(\theta) = \lim_{x \downarrow 0} \frac{\log\left(\frac{e^{x\theta}}{2} + \frac{e^{-x\theta}}{2}\right)}{x^2}$$

日中、夕に、定建を、用い、る

$$\begin{aligned}
 & \partial_x \left\{ \log \left(\frac{e^{\partial x}}{2} + \frac{e^{-\partial x}}{2} \right) \right\} \\
 &= \partial_x \left\{ \log (e^{\partial x} + e^{-\partial x}) - \log 2 \right\} \\
 &= \frac{\partial (e^{\partial x} - e^{-\partial x})}{e^{\partial x} + e^{-\partial x}}
 \end{aligned}$$

F.)

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \phi_n(\theta) \\
 &= + \lim_{x \downarrow 0} \frac{\partial (e^{\partial x} - e^{-\partial x})}{2x (e^{\partial x} + e^{-\partial x})} \\
 &= \frac{\partial t}{2} \lim_{x \downarrow 0} \frac{e^{\partial x} - e^{-\partial x}}{e^{\partial x} + e^{-\partial x}}
 \end{aligned}$$

$$x_3 - \frac{1}{2} \ln 2 \approx 0.11$$

$$\partial_x (e^{\partial x} - e^{-\partial x})$$

$$= \partial (e^{\partial x} + e^{-\partial x})$$

$$\partial_x \{ x(e^{\partial x} + e^{-\partial x}) \}$$

$$= (e^{\partial x} + e^{-\partial x}) + \partial_x (e^{\partial x} - e^{-\partial x})$$

8.2

$$\lim_{n \rightarrow \infty} \gamma_n(\theta)$$

$$= \frac{\partial t}{2} \lim_{x \rightarrow 0} \frac{\partial (e^{\partial x} + e^{-\partial x})}{(e^{\partial x} + e^{-\partial x}) + \partial_x (e^{\partial x} - e^{-\partial x})}$$

$$= \frac{\partial t}{2}$$

◦ 二項モデルの極限 $n \rightarrow \infty$ の正規分布

$$S \quad x \begin{cases} \rightarrow x \quad S(1 + \frac{\sigma}{\sqrt{n}}) \\ \rightarrow x \quad S(1 - \frac{\sigma}{\sqrt{n}}) \end{cases}$$

$$\begin{array}{c} | \qquad \qquad \qquad | \rightarrow \\ t - \frac{1}{n} \quad \xrightarrow{1 \pm \frac{\sigma}{\sqrt{n}}} \quad t \leftarrow \frac{nt}{n} \end{array}$$

$$\tilde{p} = \frac{1 + \frac{\sigma}{\sqrt{n}} - \frac{\sigma}{\sqrt{n}}}{1 + \frac{\sigma}{\sqrt{n}} - \frac{\sigma}{\sqrt{n}}}$$

$$= \frac{1 + 0 - (1 - \frac{\sigma}{\sqrt{n}})}{1 + \frac{\sigma}{\sqrt{n}} - (1 - \frac{\sigma}{\sqrt{n}})}$$

$$= \frac{\frac{\sigma}{\sqrt{n}}}{2 \frac{\sigma}{\sqrt{n}}} = \frac{1}{2}$$

$$\tilde{q} = \frac{1}{2}$$

H_{nt} : 表へ出た回数

T_{nt} : 裏へ出た回数

$$n_t = H_{nt} + T_{nt} : \text{関係式} - (1)$$

$$\text{ラ・ダ・ム・ウ・ア・ウ} \quad M_{nt} = H_{nt} - T_{nt} - (2)$$

$$(1) + (2)$$

$$n_t + M_{nt} = 2H_{nt}$$

$$(1) - (2)$$

$$n_t - M_{nt} = 2T_{nt}$$

$$\rightarrow \begin{cases} H_{nt} = \frac{1}{2}(n_t + M_{nt}) \\ T_{nt} = \frac{1}{2}(n_t - M_{nt}) \end{cases}$$

時刻 t の株価は。

$$S_n(t) = S(0) \prod_n^{H_{nt}} d_n^{T_{nt}}$$

と表す。理由は。

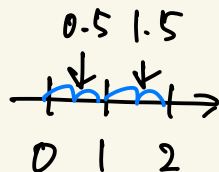
1. $n=1, t=1$

$$H_1 = 1, T_1 = 0 \text{ 以上}$$

$$\begin{aligned} S(0) &\rightarrow S(0) + \sigma S(0) \\ &= S(0) (1 + \sigma)^1 (1 - \sigma)^0 \end{aligned}$$

2. $n=2, t=2$

$$H_2 = 3, T_2 = 1$$



$$S(0) \rightarrow S(0) \left(1 + \frac{\sigma}{\sqrt{2}}\right)$$

$$\rightarrow S(0) \left(1 + \frac{\sigma}{\sqrt{2}}\right)^2 \rightarrow S(0) \left(1 + \frac{\sigma}{\sqrt{2}}\right)^3$$

$$\rightarrow S(0) \left(1 + \frac{\sigma}{\sqrt{2}}\right)^3 \left(1 - \frac{\sigma}{\sqrt{2}}\right)$$

$$\begin{aligned}
 S_n(t) &= S(0) \int_0^t \frac{H_{nt}}{S(u)} du \\
 &= S(0) \left(1 + \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt + M_{nt})} \\
 &\quad \cdot \left(1 - \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt - M_{nt})} \quad \text{--- ④}
 \end{aligned}$$

∴ 式 ④ $n \rightarrow \infty$ に近づけることを考える。

Th 3.2.2)

④ 式 は $n \rightarrow \infty$ と近づける

$$S(t) = S(0) \exp \left\{ \sigma W(t) - \frac{\sigma^2 t}{2} \right\}$$

$$\therefore \log S_n(t)$$

$$\begin{aligned}
 &= \log S(0) + \frac{1}{2}(\underline{nt} + \underline{M_{nt}}) \log \left(1 + \frac{\sigma}{\sqrt{n}}\right) \\
 &\quad + \frac{1}{2}(\underline{nt} - \underline{M_{nt}}) \log \left(1 - \frac{\sigma}{\sqrt{n}}\right)
 \end{aligned}$$

Remark

$$f(x) = \log(1+x)$$

$$f' = \frac{1}{1+x}, \quad f'' = -\frac{1}{(1+x)^2}$$

$$\begin{aligned}\rightarrow f(x) &\simeq 0 + \frac{1}{1}x - \frac{1}{2}x^2 \\ &= x - \frac{1}{2}x^2\end{aligned}$$

$$\log\left(1 + \frac{\sigma}{\sqrt{n}}\right) \simeq \frac{\sigma}{\sqrt{n}} - \frac{1}{2} \frac{\sigma^2}{n}$$

$$\log\left(1 - \frac{\sigma}{\sqrt{n}}\right) \simeq -\frac{\sigma}{\sqrt{n}} - \frac{1}{2} \frac{\sigma^2}{n} \quad \text{f.)}$$

$$\log S_n(t) \simeq \log S(0)$$

$$+ \frac{1}{2} (nt + M_{nt}) \left\{ \frac{\sigma}{\sqrt{n}} - \frac{1}{2} \frac{\sigma^2}{n} \right\}$$

$$+ \frac{1}{2} (nt - M_{nt}) \left\{ -\frac{\sigma}{\sqrt{n}} - \frac{1}{2} \frac{\sigma^2}{n} \right\}$$

$$= \log S(0) + M_{nt} \frac{\sigma}{\sqrt{n}} + nt \left(-\frac{\sigma^2}{2n} \right)$$

$$= \log P(0) - \frac{\sigma^2 t}{2} + M_{nt} \frac{\sigma}{\sqrt{n}}$$

Remark

$$| \overline{W}_t^{(n)} = \frac{1}{\sqrt{n}} M_{nt} |$$

$$= \log P(0) - \frac{\sigma^2 t}{2} + \sigma \overline{W}_t^{(n)}$$

Remark 中心极限定理

$$| \overline{W}_t^{(n)} \xrightarrow{d} N(0, t) |$$

$$\lim_{n \rightarrow \infty} (\log S_n(t))$$

$$= \log P(0) - \frac{\sigma^2 t}{2} + \sigma \overline{W}(t)$$

$$\rightarrow \lim_{n \rightarrow \infty} P_n(t) = P(0) \exp\left(-\frac{\sigma^2 t}{2} + \sigma \overline{W}(t)\right)$$

Remark 对数正态分布

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$$\text{例} \quad Y = \log X \sim N(\mu, \sigma^2)$$

期望值 $X = e^Y \sim \text{LN}(\mu, \sigma^2)$ 且

$$E[X] = E[e^{N(\mu, \sigma^2)}]$$

$$= \exp\left(\mu + \frac{\sigma^2}{2}\right)$$

中央值

$$P(X < c) = P(e^{N(\mu, \sigma^2)} < c)$$

$$= P(N(\mu, \sigma^2) < \log c)$$

$$= P(N(0, 1) < \frac{\log c - \mu}{\sigma})$$

$$= \Phi\left(\frac{\log c - \mu}{\sigma}\right) = 0.5$$

$$\rightarrow \frac{\log c - \mu}{\sigma} = 0$$

$$\log c = \mu \rightarrow c = e^\mu$$

最頻偏

$$\log f(x) \propto -\log x - \frac{(\log x - \mu)^2}{2\sigma^2}$$

$$\partial_x (\log f(x))$$

$$= -\frac{1}{x} - \frac{1}{\sigma^2} \frac{1}{x} (\log x - \mu) = 0$$

$$-\frac{1}{x} = \frac{1}{x\sigma^2} (\log x - \mu)$$

$$-\sigma^2 = \log x - \mu$$

$$\rightarrow x_{\text{mode}} = \exp(\mu - \sigma^2)$$

$$\vec{x} = (x_1, x_2, \dots, x_n)$$

$$L(\mu, \sigma^2) = \prod_{i=1}^n f(x_i)$$

$$l(\mu, \sigma^2) = \sum_i \log f(x_i)$$

$$= \sum_i \left\{ -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma^2 - \log x_i - \frac{(\log x_i - \mu)^2}{2\sigma^2} \right\}$$

$$\alpha_{\mu, \sigma^2} = \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (\log x_i - \mu)^2$$

$$\partial_{\mu} (l(\mu, \sigma^2))$$

$$= \frac{1}{\sigma^2} \sum_i (\log x_i - \mu) = 0$$

$$\hat{\mu} = \overline{\log x}$$

◦ 7"うり: 重要.

各種計算

$$\cdot E[W(s)W(t)] \quad (t > s)$$

$$= E[(W(t) - W(s) + W(s))W(s)]$$

$$= E[W(t) - W(s)]E[W(s)]$$

$$+ E[W^2(s)] = s.$$

◦ 区間係数

$$\text{例: } u_3 W(t_3) + u_2 W(t_2) + u_1 W(t_1)$$

$$= u_3 (W(t_3) - W(t_2))$$

$$+ (u_3 + u_2) (W(t_2) - W(t_1))$$

$$+ (u_3 + u_2 + u_1) W(t_1)$$

7"系 2.2 を 利用し,

$$\vec{W} = [W(t_1) \ W(t_2) \ \dots \ W(t_m)]$$

の $\mathbb{R}$ -valued 母関数。

$$\varphi(u_1, u_2, \dots, u_m)$$

$$= E \left[\exp \left\{ \sum_{i=1}^m u_i W(t_i) \right\} \right]$$

$$= E \left[\exp \int u_m (W(t_m) - W(t_{m-1})) \right.$$

$$+ (u_m + u_{m-1}) (W(t_{m-1}) - W(t_{m-2}))$$

$$+ (u_m + u_{m-1} + u_{m-2}) (W(t_{m-2}) - W(t_{m-3}))$$

$$+ \dots + (u_m + u_{m-1} + u_{m-2} + \dots + u_1) W(t_1) \Big]]$$

$$= E \left[\exp (u_m (W(t_m) - W(t_{m-1}))) \right]$$

$$\cdot E \left[\exp \left\{ (u_m + u_{m-1}) (W(t_{m-1}) - W(t_{m-2})) \right\} \right]$$

$$\dots E \left[\exp \left\{ (u_m + u_{m-1} + \dots + u_1) W(t_1) \right\} \right]$$

$$= \exp\left(\frac{1}{2} u_m^2 (t_m - t_{m-1})\right)$$

$$\exp\left(\frac{1}{2} (u_m + u_{m-1})^2 (t_{m-1} - t_{m-2})\right)$$

$$\dots \exp\left(\frac{1}{2} (u_m + u_{m-1} + \dots + u_1)^2 t_1\right)$$

0 2次変分

$$||\pi|| = \max_j (t_{j+1} - t_j)$$

(3.4.1 : 1次変分)

$$FU_T(f) = \lim_{||\pi|| \rightarrow 0} \sum_{j=0}^{n-1} |f(t_{j+1}) - f(t_j)|$$

∴ ?

$$\pi = \{t_0, t_1, \dots, t_n\}$$

Remark : 平均値の定理)

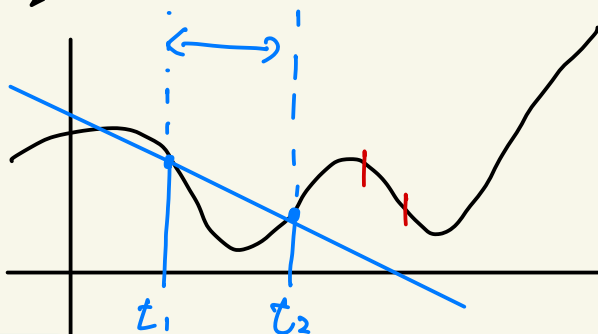
各部分区間 $[t_j, t_{j+1}]$ に対し

次の点 t_j^* が存在する

$$\frac{f(t_{j+1}) - f(t_j)}{t_{j+1} - t_j} = f'(t_j^*)$$

例 2

この区間の中に同じ値をとる点がある



平均値の定理に於て式(1)

$$f(t_{j+1}) - f(t_j) = f'(t_j^*) (t_{j+1} - t_j)$$

F.)

$$FV_T(f) = \lim_{\|T\| \rightarrow 0} \sum_{j=0}^{n-1} |f'(t_j^*)| (t_{j+1} - t_j)$$

Remark : (1-2) 和)

$$\left| \sum f(\xi_i) \Delta x_i \right| \leftrightarrow \int f(x) dx$$

$$FV_T(f) = \int_0^T dt f'(t)$$

○ 2次元変分.

def

$f(t)$ を $0 < t < T$ で定義された関数
とし.

$$[f, f](T) = \lim_{\|\pi\| \rightarrow 0} \sum_{j=0}^{n-1} [f(t_{j+1}) - f(t_j)]^2$$

※ 平均値の定理が利用できるように $f(t)$
は 2次元変分は 0 とする。

一方、ブラウン運動の経路は.

時間に対し微分可能ではない.

とし、以下、定理が成り立つ

Th 3.4.3

W がブラウン運動ならば

$$[W, W](T) = T$$

∴ 可算得)

$$Q_{\pi} = \sum_{j=0}^{n-1} (\overline{W}(t_{j+1}) - \overline{W}(t_j))^2$$

期得值

$$E[\overline{W}(t_{j+1}) - \overline{W}(t_j)]^2]$$

$$= t_{j+1} + t_j - 2t_j = t_{j+1} - t_j$$

Remark

$$E[\overline{W}(t+s)\overline{W}(t)]$$

$$= E[(\overline{W}(t+s) - \overline{W}(t) + \overline{W}(t))\overline{W}(t)]$$

$$= 0 + t.$$

5.2

$$E[Q_{\pi}]$$

$$= \sum_{j=0}^{n-1} (t_{j+1} - t_j) = t_n - t_0 = T$$

· 分散

$$\text{Var} [(W(t_{j+1}) - W(t_j))^2]$$

$$= E \left[\left\{ (W(t_{j+1}) - W(t_j))^2 - \underbrace{E[(W(t_{j+1}) - W(t_j))^2]}_{t_{j+1} - t_j} \right\}^2 \right]$$

$$= E \left[\left\{ (W(t_{j+1}) - W(t_j))^2 - (t_{j+1} - t_j) \right\}^2 \right]$$

$$= E \left[(W(t_{j+1}) - W(t_j))^4 \right]$$

$$- 2(t_{j+1} - t_j) E \left[(W(t_{j+1}) - W(t_j))^2 \right]$$

$$+ (t_{j+1} - t_j)^2$$

= 2.2:

$$X \sim N(0, \sigma^2) \text{ on } \mathbb{R} \text{ with } \mu_X = 0$$

$$M_X(\theta) = \exp\left(\frac{\theta^2 \sigma^2}{2}\right)$$

$$M_X'(\theta) = \sigma^2 \theta \exp\left(\frac{\theta^2 \sigma^2}{2}\right) = \sigma^2 \theta M_X$$

$$M_X''(\theta) = \sigma^2 M_X(\theta)$$

$$+ \sigma^4 \theta^2 M_X(\theta)$$

$$M_X^{(3)}(\theta) = \sigma^4 \theta M_X(\theta)$$

$$+ 2\sigma^4 \theta M_X(\theta)$$

$$+ \sigma^4 \theta^2 \cdot \sigma^2 \theta M_X(\theta)$$

$$M_X^{(4)}(\theta) = \sigma^4 M_X(\theta) + \sigma^4 \theta \cdot \sigma^2 \theta M_X$$

$$+ 2\sigma^4 M_X(\theta) + 2\sigma^4 \theta \cdot \sigma^2 \theta M_X$$

$$+ 3\sigma^6 \theta^2 M_X(\theta) + \dots$$

$$\Rightarrow M_X^{(4)}(\theta) = \sigma^4 + 2\sigma^4 = 3\sigma^4 \quad \square$$

$$\text{for } W(t_{j+1}) - W(t_j) \sim N(0, t_{j+1} - t_j)$$

$$E[(W(t_{j+1}) - W(t_j))^2]$$

$$= 3(t_{j+1} - t_j)^2$$

for

$$\text{Var} \sum (W(t_{j+1}) - W(t_j))^2]$$

$$= 3(t_{j+1} - t_j)^2 - 2(t_{j+1} - t_j)^2$$

$$+ (t_{j+1} - t_j)^2$$

$$= 2(t_{j+1} - t_j)^2$$

$$\rightarrow \text{Var} [Q_n]$$

$$= \sum_{j=0}^{n-1} 2(t_{j+1} - t_j)^2$$

$$\leq \sum_{j=0}^{n-1} 2 \|\pi\| (t_{j+1} - t_j)$$

$$= 2||\pi|| \cdot T$$

$$\longrightarrow \text{Var}[Q\pi] = 0$$

$$||\pi|| \rightarrow 0$$

期待値 $E[Q\pi]$ は

経路に依存しない T

□

・ 幾何ブラウン運動のボラティリティ

幾何ブラウン運動

$$S(t) = S(0) \exp \left\{ \sigma W(t) + \left(\alpha - \frac{\sigma^2}{2} \right) t \right\}$$

ここで、この経路からボラティリティ σ

を推定するために σ^2 のように

ブラウン運動の2次変分を用いる。

○ log Return を考える

$$\log \frac{S(t_{j+1})}{S(t_j)}$$

$$= \log \left\{ \exp \left(\sigma W(t_{j+1}) + \left(\alpha - \frac{\sigma^2}{2} \right) t_{j+1} \right) \right. \\ \left. - \sigma W(t_j) - \left(\alpha - \frac{\sigma^2}{2} \right) t_j \right\}$$

$$= \sigma(\overline{W}(t_{j+1}) - \overline{W}(t_j)) \\ + \left(\alpha - \frac{\sigma^2}{2}\right)(t_{j+1} - t_j)$$

次に 実際 ポロノミヤ は

$$\sum_{j=0}^{n-1} \left(\log \frac{S(t_{j+1})}{S(t_j)} \right)$$

$$= \sigma^2 \sum_{j=0}^{n-1} (\overline{W}(t_{j+1}) - \overline{W}(t_j))^2 \\ + 2\sigma\left(\alpha - \frac{\sigma^2}{2}\right) \sum_{j=0}^{n-1} (\overline{W}(t_{j+1}) - \overline{W}(t_j)) \\ (t_{j+1} - t_j)$$

$$+ \left(\alpha - \frac{\sigma^2}{2}\right)^2 \sum_{j=0}^{n-1} (t_{j+1} - t_j)^2$$

$$||\Pi|| \rightarrow 0 \quad \varepsilon_1 \varepsilon_2 \varepsilon_3$$

$$\text{第1項は } \sigma^2(T_2 - T_1) \wedge$$

$$\text{第2項は } 0$$

$$\text{第3項は } 0$$

$$\Rightarrow \frac{1}{T_2 - T_1} \sum_{j=0}^{m-1} \left(\log \frac{S'(t_{j+1})}{S(t_j)} \right)^2 \approx \sigma^2$$

到達時間分布

・ 指数 2 次元 $\tau = \tau' - 1$

$$Z(t) = \exp \left\{ \sigma W(t) - \frac{\sigma^2}{2} t \right\}$$

・ 指数 2 次元 $\tau = \tau' - 1$

$$\therefore E[Z(t) | F(s)] \sim N(0, t-s)$$

$$= E \left[\exp \left\{ \sigma (W(t) - W(s)) \right\} \right]$$

$$\exp \left\{ \sigma W(s) - \frac{\sigma^2}{2} t \right\} | F(s)]$$

$$= \exp \left(\sigma W(s) - \frac{\sigma^2}{2} t \right)$$

$$\exp \left(\frac{\sigma^2}{2} (t-s) \right)$$

$$= \exp \left(\sigma W(s) - \frac{\sigma^2}{2} s \right)$$

$$= Z(s)$$

□

?? $E[Z(t)] = Z(0)$ or ?

$$Z(0) = \exp\left(\sigma W(0) - \frac{\sigma^2}{2} 0\right) = 1$$

$$E[Z(t)] = \exp\left(-\frac{\sigma^2}{2} t\right) E[\exp(\sigma W(t))]$$

$N(0, t)$
↙

↙

$$\exp\left(\frac{\sigma^2}{2} t\right)$$

$$= 1$$

$$\begin{cases} \tau_m \equiv \min \{ t \geq 0 ; W(t) = m \} \\ t \wedge \tau_m : t \leq \tau_m \text{ のとき } \dots \text{ (最小値)} \end{cases}$$

と定義する。

$$Z(0) = 1 \quad \text{任意の } t \text{ に対して成り立つ}$$

$$= E[Z(t \wedge \tau_m)]$$

$$= E\left[\exp\left(\sigma W(t \wedge \tau_m) - \frac{\sigma^2}{2}(t \wedge \tau_m)\right)\right]$$

$\because \sigma > 0, m > 0 \text{ として}$
計算して...

$$t \leq \tau_m \text{ ならば } W(t) \leq m$$

ゆえに

$$0 \leq \exp(\sigma W(t \wedge \tau_m)) \leq e^{\sigma m}$$

が成り立つ

$$T_m < \infty \quad (: \text{finite } T_m \text{ exists})$$

$$\exp\left(-\frac{\sigma^2}{2}(t \wedge T_m)\right)$$

$$\xrightarrow{t \rightarrow \infty} \exp\left(-\frac{\sigma^2}{2} T_m\right)$$

$$T_m = \infty \quad (: \text{no finite } T_m \text{ exists})$$

$$\exp\left(-\frac{\sigma^2}{2}(t \wedge T_m)\right)$$

$$\xrightarrow{t \rightarrow \infty} \exp\left(-\frac{\sigma^2}{2} t\right) = 0$$

~~~~~

$$\lim_{t \rightarrow \infty} \exp\left(-\frac{\sigma^2}{2}(t \wedge T_m)\right)$$

$$= \mathbb{1}[T_m < \infty] \exp\left(-\frac{\sigma^2}{2} T_m\right)$$

77.  $T_m < \infty$  a.s.  $\frac{1}{\sqrt{2}}$ .

$$\exp(\sigma W(t \wedge T_m))$$

$$\xrightarrow{t \rightarrow \infty} \exp(\sigma \underbrace{W(T_m)}_m)$$

$$T_m = \infty \text{ a.s. } \frac{1}{\sqrt{2}}$$

$$\exp(\sigma W(t \wedge T_m))$$

$$\xrightarrow{t \rightarrow \infty} \text{exists } e^{\sigma m} \text{ is finite.}$$

$$W \in \mathcal{D}^{1,2}$$

$$\lim_{t \rightarrow \infty} \exp \left\{ \sigma W(t \wedge T_m) - \frac{\sigma^2}{2} (t \wedge T_m) \right\}$$

$$= \mathbb{1}[T_m < \infty] \exp \left( \sigma m - \frac{\sigma^2}{2} T_m \right)$$

計 算 7:

$$E\left[\exp\left\{\sigma W(t \wedge \tau_m) - \frac{\sigma^2}{2}(t \wedge \tau_m)\right\}\right]$$

$$= E\left[\mathbb{1}(\tau_m < \infty) \exp\left(\sigma m - \frac{\sigma^2}{2} m\right)\right]$$

$$= 1$$

$$\Rightarrow E\left[\mathbb{1}(\tau_m < \infty) \exp\left(-\frac{\sigma^2}{2} \tau_m\right)\right]$$

$$= e^{-\sigma m}$$

$\tau_m$  の 1 次元 Brownian motion の 1 次元

$$E\left[\exp\left(-\frac{\sigma^2}{2} \tau_m\right)\right] = e^{-\sigma m}$$

## Th 3.6.2

$m \in \mathbb{R}$  に対して  $W_m$  のブラウン運動の到達時間  $\tau_m$  はほとんど有限であり、

$\tau_m$  の分布のラプラス変換は  $\alpha > 0$  として、

$$E[e^{-\alpha \tau_m}] = \exp(-|m| \sqrt{2\alpha})$$

と表す

$\therefore$  先の式に  $\alpha = \frac{\sigma^2}{2}$  とおくと

$$\sigma = \sqrt{2\alpha}$$

# ○ 鏡像原理

(目標)

(正のレベル  $m$  と正の時刻  $t$  を固定し、  
時刻  $t$  以前に、レベル  $m$  に到達する  
ブラウン運動の経路を数える)

鏡像等式

$$\Pr[T_m \leq t, W(t) \leq w]$$

時刻  $t$  以前にレベル  $m$  に到達し

時刻  $t$  に  $w$  を下回る位置

$$= \Pr[W(t) \geq 2m - w]$$

Ex 3.7.1)

$$\begin{aligned} & \Pr[T_m \leq t] \\ &= \frac{2}{\sqrt{2\pi}} \int_{\frac{|m|}{\sqrt{t}}}^{\infty} e^{-\frac{y^2}{2}} dy \quad (t > 0) \end{aligned}$$

2.2.2).

$$\begin{aligned} f_{T_m}(t) &= \frac{d}{dt} \Pr[T_m \leq t] \\ &= \frac{1}{\sqrt{2\pi t}} \frac{|m|}{t} e^{-\frac{m^2}{2t}} \end{aligned}$$

∴ 鏡像等式 8.1)  $W = m \leq 12$

$$\Pr[T_m \leq t, W(t) \leq m]$$

$$= \Pr[W(t) \geq m]$$

※7. 以下を証明せよ

$$\Pr[\tau_m \leq t, \overline{W}(t) \geq m]$$

↑  
独立

$$= \Pr[\overline{W}(t) \geq m]$$

※8

$$\Pr[\tau_m \leq t]$$

$$= \Pr[\tau_m \leq t, \overline{W}(t) \geq m]$$

$$+ \Pr[\tau_m \leq t, \overline{W}(t) \leq m]$$

$$= 2 \Pr[\overline{W}(t) \geq m]$$

$$= 2 \frac{1}{\sqrt{2\pi t}} \int_m^\infty dw e^{-\frac{w^2}{2t}}$$

$$y = \frac{w}{\sqrt{t}}$$

$$dy = \frac{dw}{\sqrt{t}}$$

$$\Pr[\tau_m \leq t]$$

$$= \frac{2}{\sqrt{2\pi t}} \int_{\frac{m}{\sqrt{t}}}^{\infty} dy \cdot \exp\left(-\frac{y^2}{2}\right)$$

Remark

(B5) ライプニッツの公式:  $\theta$  に関して微分可能で  $x$  に関して積分可能な関数  $f(x, \theta)$  に対して

$$\begin{aligned} \frac{d}{d\theta} \int_{g(\theta)}^{h(\theta)} f(x, \theta) dx &= f(h(\theta), \theta) \frac{dh(\theta)}{d\theta} - f(g(\theta), \theta) \frac{dg(\theta)}{d\theta} \\ &\quad + \int_{g(\theta)}^{h(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx. \end{aligned}$$

となる.

$$\begin{aligned} \left(\frac{1}{\sqrt{t}}\right)' &= \left(t^{-\frac{1}{2}}\right)' \\ &= -\frac{1}{2} t^{-\frac{3}{2}} = -\frac{1}{2t\sqrt{t}} \end{aligned}$$

(B5) ライブニッツの公式:  $\theta$  に関して微分可能で  $x$  に関して積分可能な関数  $f(x, \theta)$  に対して

$$\frac{d}{d\theta} \int_{g(\theta)}^{h(\theta)} f(x, \theta) dx = f(h(\theta), \theta) \frac{dh(\theta)}{d\theta} - f(g(\theta), \theta) \frac{dg(\theta)}{d\theta} + \int_{g(\theta)}^{h(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx.$$

となる.

$$\Pr[\tau_m \leq t] = \frac{2}{\sqrt{2\pi}} \int_{\frac{|m|}{\sqrt{t}}}^{\infty} e^{-\frac{y^2}{2}} dy$$

$$\begin{aligned} \frac{d}{dt} \Pr[\tau_m \leq t] \\ = \sqrt{\frac{2}{\pi}} \frac{m}{2t\sqrt{t}} e^{-\frac{m^2}{2t}} \end{aligned}$$

$$= \frac{m}{t\sqrt{2\pi t}} e^{-\frac{m^2}{2t}}$$

Remark 逆ガウス分布

Q26 (2点)

$$f(x|\mu, \lambda) \\ = \left( \frac{\lambda}{2\pi x^3} \right)^{\frac{1}{2}} \exp \left( - \frac{\lambda(x-\mu)^2}{2\mu^2 x} \right)$$

・ ブラウニアン運動と  $Z$  の最大値の分布.

$$M(t) \equiv \max_{0 \leq s \leq t} W(s)$$

鏡像等式より

$$\Pr[T_m \leq t, W(t) \leq w]$$

$$= \Pr[W(t) \geq 2m - w]$$

\*  
 $\rightarrow \Pr[M(t) \geq m, W(t) \leq w]$

$$= \Pr[W(t) \geq 2m - w]$$

Th 3.7.3

$(M(t), W(t))$  a 2D 密度函数

$$f(m, w) = \frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

$$P_r[M(t) \geq m, W(t) \leq w]$$

$$= \int_m^\infty dx \int_{-\infty}^w dy f(x, y) \quad - ①$$

$$P_r[W(t) \geq 2m-w]$$

$$= \frac{1}{\sqrt{2\pi t}} \int_{2m-w}^\infty dz e^{-\frac{z^2}{2t}} \quad - ②$$

$$① = ② \quad \text{证 (*)} \quad ?$$

证: 用 2 变量

$$- \int_{-\infty}^w f(m, y) dy$$

$$= - \frac{1}{\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

w 7. 22.2.1

$$- f(m, w)$$

$$= - \frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

□.

Ex:

$M(t) | W(t)$  is i.i.d.

$$f(m|w) = \frac{f_{M(t), W(t)}(m, w)}{f_{W(t)}(w)}$$

$$= \frac{\frac{2(2m-w)}{t\sqrt{2\pi t}}}{\frac{1}{\sqrt{2\pi t}}} \exp \left\{ - \frac{(2m-w)^2}{2t} + \frac{w^2}{2t} \right\}$$

$$= - \frac{1}{2t} (4m^2 - 4mw + w^2 - w^2)$$

$$= - \frac{2m}{t} (m - w)$$

$$\Rightarrow f(m(w))$$

$$= \frac{2(2m-w)}{t} \exp \left\{ - \frac{2m(m-w)}{t} \right\}$$

□